

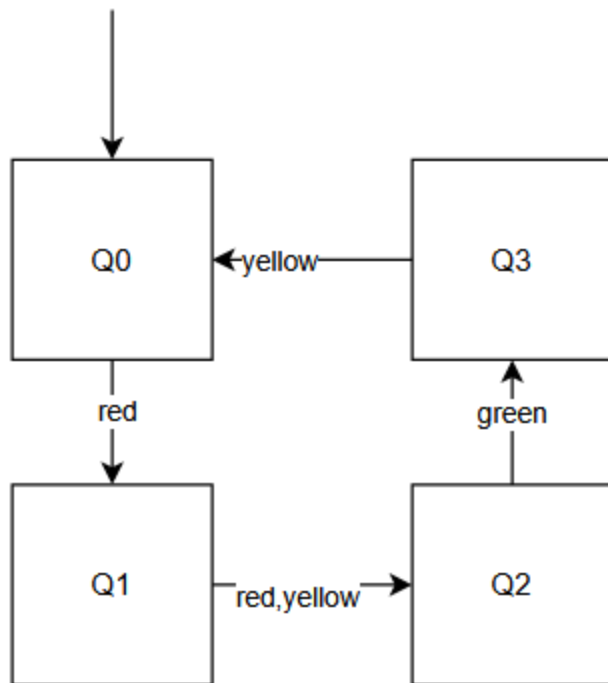
CPS I — Exercise Sheet 9

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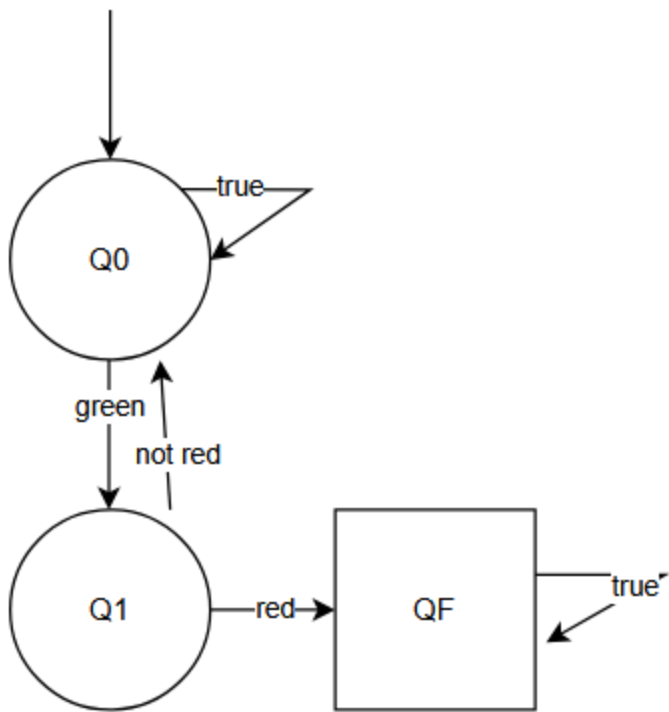
Exercise 1: Model Checking

1. A_T

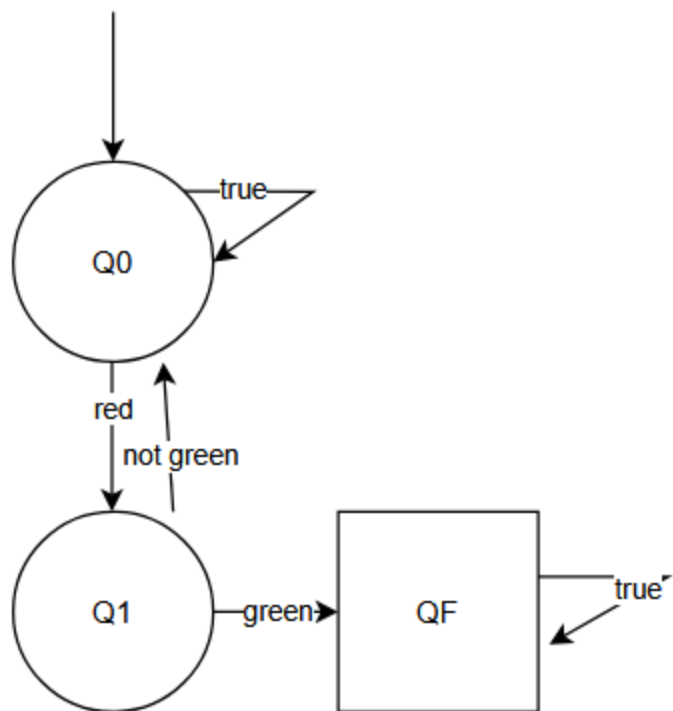


2. Bad Prefixes

A_{P1}



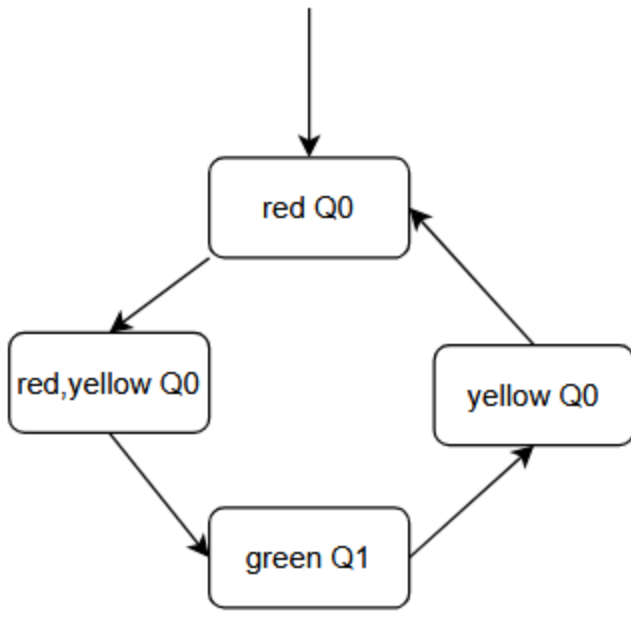
A_{P2}



3. Intersection

Intersection of A_T and A_{P1}

no accepting state can be reached



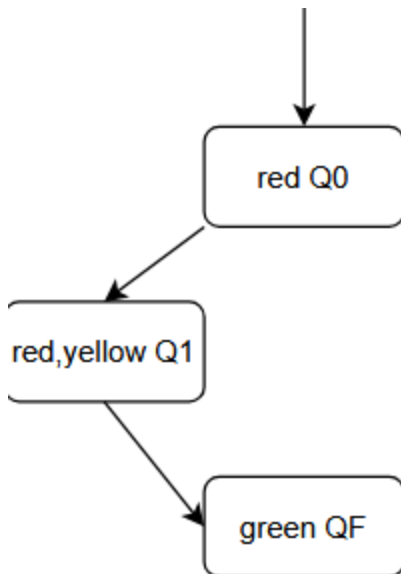
Intersection of A_T and A_{P2}

Is accepting state can be reached

$$\mathcal{L}(A_T \cap A_{P_2}) \neq \emptyset$$

So TS doesn't satisfy P2

Counter Example: $\{\text{red}\}\{\text{red,yellow}\}\{\text{green}\}$



Exercise 2: Safety & Liveness

- 1
 - Safety $E = \{\sigma \in \Sigma^\omega \mid a \in A_0\}$
 - Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, a \in A_i\}$
- 2

- Safety $E = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, a \notin A_i\}$
- Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists^\infty i \geq 0, a \in A_i\}$
- 3
 - Safety $E = \{\sigma \in \Sigma^\omega \mid b \in A_0\}$
 - Liveness $E = \{\sigma \in \Sigma^\omega\}$
- 4
 - Safety: No, because every trace violates safety property has a bad prefix
 - Liveness $E = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, a \in A_i\}$

Exercise 3: LT Properties

- $P_1 = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, a \in A_i \vee b \in A_i\}$
 - Invariant
 - Safety
 - not Liveness
- $P_2 = \{\sigma \in \Sigma^\omega \mid \exists i \geq 0, (a \in A_i \wedge \forall j \neq i, a \notin A_j)\} \cup \{\sigma \in \Sigma^\omega \mid \forall i \geq 0, b \notin A_i\}$
 - not Invariant
 - not Safety
 - not Liveness
- $P_3 = \{\sigma \in \Sigma^\omega \mid \forall i \geq 0 (a \in A_i) \implies (b \notin A_{i+1})\}$
 - not Invariant
 - Safety
 - not Liveness